Niharika Vullaganti

Precision Agriculture | Soil Health | Hyperspectral Sensing | Artificial Intelligence | Robotics

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LinkedIn Google Scholar ResearchGate X

Personal Statement

I am passionate about driving sustainability in agriculture through precision agriculture technologies. My research integrates hyperspectral sensing artificial intelligence, and robotics for site-specific nutrient management. Looking ahead, I aim to continue and expand my research in the areas of site-specific input management and variable rate technology in agriculture. My goal is to serve multiple stakeholders involved in food production system.

Research Interests

Precision agriculture, proximal and remote sensing, artificial intelligence, site-specific input management, robotics and drones in agriculture, data-driven agronomy, sustainable intensification.

Education

Ph.D. Natural Resource Management (CGPA: 4.00/4.00) Jan 2024 – Expected May 2027

North Dakota State University, ND, USA | *Advisor:* Dr. Xin (Rex) Sun

Graduate Certificate in Big Data Applied Statistical Analysis (CGPA: 4.00/4.00) May 2024 – August 2025

North Dakota State University, ND, USA

M.Sc. Soil Science (CGPA: 8.15/10.00) Oct 2020 – Nov 2022

Punjab Agricultural University, Punjab, India | *Advisor:* Dr. Bharat Bhushan Vashisht

B.Sc. (Hons) Agriculture (CGPA: 8.77/10.00) Aug 2016 – Aug 2020

Acharya N.G. Ranga Agricultural University, AP, India | *Advisor:* Dr. Ravi Babu

Research & Professional Experience

Graduate research assistant Jan 2024 – Present

Department of Agricultural and Biosystems Engineering, NDSU, ND, USA
Advisor: Dr. Xin (Rex) Sun

- Working on hyperspectral imaging and deep learning for soil nutrient prediction and integrating them onto a robot for real-time soil nutrient assessment, in collaboration with engineers, agronomists and soil scientists.
- Developed a low-cost IoT-based smart irrigation prototype for real-time irrigation scheduling.
- Supported University of Minnesota collaborators with pXRF field data collection for a sulfur stress study in corn.

Laboratory assistant Jan 2023 – Oct 2023

Soil Testing Laboratory, Government of AP, India | *Advisor:* Mr. Rama Mohan

Analyzed 1,000+ soil samples for nutrient and quality parameters to prepare region-specific soil health cards.

Graduate research assistant Mar 2021 – Nov 2022

Punjab Remote Sensing Centre, Punjab, India | *Advisor:* Dr. Anil Sood

Used satellite imagery and GIS tools (Google Earth Engine, Sentinel, ArcGIS, QGIS) to map evapotranspiration, irrigation demand, and cropping patterns.

Undergraduate research intern June 2019 – Aug 2019

Kansas State University, KS, USA | *Professor:* Dr. Charles W. Rice

Studied the role of soil microbial communities in greenhouse gas mitigation and engaged in interdisciplinary lab rotations, visited genomics facilities, precision agriculture labs, and the Wheat Innovation Center.

Agricultural intern Dec 2019 – Feb 2020

Syngenta, AP, India | *Advisor:* Mr. Kiran Kumar

Supported hybrid corn seed production across 1,500 acres.

Skill Set

- **Programming & Data analysis:** Python (NumPy, Pandas, Matplotlib, TensorFlow, scikit-learn), R (data wrangling, statistical computing, and visualization with ggplot2 and plotly), MATLAB (descriptive statistics, data analysis, and visualization).
- **Statistics & Machine learning:** Deep learning (multitask neural networks, CNN/ ResNet/ DenseNet for spectral data), applied statistical machine learning (regression, classification, cross-validation, and model evaluation), exploratory data analysis, and big data visualization.
- **Sensing technologies & GIS tools:** Hyperspectral imaging, remote sensing, portable X-ray fluorescence (pXRF), Google Earth Engine, ERDAS Imagine, ArcGIS Pro and QGIS.
- **Field experimentation & laboratory techniques:** Field experiment planning, data collection, and soil sampling, soil testing including macro- and micronutrient analysis, textural analysis, pH and EC.
- **Others:** Operating drones and robotic systems; 3D printing; LaTeX for scientific writing, reports.

Publications

1. **Vullaganti, N.**, Ram, B., Zhang, X., Aderholdt, W., Overby, P., & Sun, X. (2026). AI-Augmented hyperspectral soil sensing: predictive modeling of nitrogen and phosphorus using neural architecture search. Precision Agriculture, 27(1), 21.
2. **Vullaganti, N.**, Vashisht, B. B., Kaur, H., & Arora, M. (2026). Satellite-Based Spatiotemporal Variability and Mapping the Irrigation Water Requirements of Wheat Crop in Ludhiana District, Punjab. Indian Journal of Soil Conservation, 54(2), 119-129.
3. **Vullaganti, N.,** Ram, B. G., & Sun, X. (2025). Precision agriculture technologies for soil site-specific nutrient management: A comprehensive review. Artificial Intelligence in Agriculture, 15(2), 147–161.
4. **Vullaganti, N.**, Zhang, X., Aderholdt, W., Overby, P., & Sun, X. (2026). Simultaneous hyperspectral classification of soil nitrogen and phosphorus using multitask deep learning across VNIR and SWIR spectral regions. Smart Agricultural Technology (Under revision, October 2026).
5. **Vullaganti, N.**, Kumar, T., & Verma, P. K (2023). Emerging Technologies for Water Management in Agriculture. Novel Approaches in Agronomy, Volume 2, 62.
6. Zhang, X., **Vullaganti, N.,** Aderholdt, W., Overby, P., & Sun, X. (2026). Systematic comparison of band selection strategies and sensor configurations for soil moisture classification using UGV-mounted hyperspectral imagery. Smart Agricultural Technology, 102170.
7. Zhang, X., Ram, B., **Vullaganti, N.**, Aderholdt, W., Overby, P., & Sun, X. (2025). Soil Moisture Classification Using Hyperspectral Imaging and Deep Learning Models on Ground Robot Vehicles. Smart Agricultural Technology, 101413.
8. Yadav, M., Vashisht, B. B., **Vullaganti, N.**, Jalota, S. K., Yadav, S. L., Singh, G., Kumar, A., & Kothiyal, S. (2024). IoT-Enabled Unmanned Aerial Vehicle: An Emerging Trend in Precision Farming. Artificial Intelligence and Smart Agriculture: Technology and Applications, 271-292.
9. Yadav, M., Vashisht, B. B., **Vullaganti, N.**, Kumar, P., Jalota, S. K., Kumar, A., & Kaushik, P. (2024). UAV-enabled approaches for irrigation scheduling and water body characterization. Agricultural Water Management, 304, 109091.
10. Kumar, T., Poomiyan, S., **Vullaganti, N.**, & Preeti (2023). Digitization and Data Analytics in Agriculture. Recent Approaches in Agriculture, Volume 2, 51.

Grant Writing Experience

- **Contributor**, "Autonomous Soil Health Monitoring: Edge AI-Enhanced UGV Platform for In-Field Sensing and Sampling," USDA NIFA Agriculture and Food Research Initiative (AFRI), Engineering for Precision Crop and Water Management (A1551). Submitted 2025; resubmission in preparation.

- PI: Dr. Xin Sun (Agricultural and Biosystems Engineering); Co-PIs: Dr. Carlos Pires (School of Natural Resource Sciences) and Dr. Laxmi Prasad (Agricultural and Biosystems Engineering), North Dakota State University.
- Contributed to conceptualization, drafting and revising the proposal, gained experience in collaboration with multiple PIs, budget justification, and planning project timeline.

Awards and Honors

- **Graduate Student Travel Award** of $750 by ASA-CSSA-SSSA and Agronomic Science Foundation to attend and present my research at CANVAS 2026 Annual Meeting, Portland, OR.
- **1st place in Digital Agriculture Hackathon, ASABE 2026**. Project: Precision cattle management using a low-cost Ultra-Wide-Band ear tag system and a digital twin platform for real-time localization and immediate intervention.
- **1st place in NDSU AgTech Hackathon, 2026** held by NDSU, NSF AgTech Engine North Dakota, and Bayer. Project: UAV-based subplot detection, weed detection, crop counting, and biomass estimation tasks.
- **Finalist, ASABE Student Oral Presentation Competition**, CSABE/ASABE Annual International Meeting, Toronto, 2025. Awarded $250 prize.
- **NDSU Possibility Fellow**, 2025 – Selected by the Center for Entrepreneurship and Family Business; awarded **$5,000** to develop an AI-powered indoor farming startup, 'HydroVerdant – A Smart Hydroponic Kitchen Garden'.
- **Killilea Possibility Fellow**, 2025–2026 – Named to honor the donors of the NDSU Possibility Fellowship.
- Qualified **University Grants Commission-National Eligibility Test (UGC-NET)**, obtaining eligibility for Assistant Professorship in Environmental Sciences in India.
- Awarded **ICAR-National Talent Scholarship**, during master's program.
- **University Merit Certificate** for outstanding academic performance during master's program.

Professional Affiliations

- Soil Science Society of America (SSSA), Member, 2024–present.
- American Society of Agronomy (ASA), Member, 2024–present.
- Crop Science Society of America (CSSA), Member, 2024–present.
- American Society of Agricultural and Biological Engineers (ASABE), Member, 2024–present.
- Association for Agricultural, Biological & Food Engineers of Indian Origin (AABFEIO), an ASABE affiliate, Member.
- Indian Society of Soil Science (ISSS), Member, 2020–2024.

Conferences and Presentations

- **Vullaganti, N.**[$], Ram, B. G., Zhang, X., Pires, C. B., Alderholdt, W., Overby, P., Sun, X. Hyperspectral Imaging and Deep Learning for Rapid Assessment of Soil Nutrient Status Using Ground Robotic Vehicles. 2026 CANVAS, November 1–4. Portland, OR. [Poster to be presented].
- **Vullaganti, N.**[$], Zhang, X., Pires, C. B., Alderholdt, W., Overby, P., Sun, X. Simultaneous Hyperspectral Classification of Soil Nitrogen and Phosphorus Using Multitask Deep Learning Across VNIR and SWIR Spectral Regions. 2026 ASABE International Meeting, July 12–15. Indianapolis, IN. [Oral].
- Zhang, X.[$], **Vullaganti, N.**, Alderholdt, W., Overby, P., Sun, X. Systematic Comparison of Band Selection Strategies and Sensor Configurations for Soil Moisture Classification Using UGV - Mounted Hyperspectral Imagery. 2026 ASABE International Meeting, July 12–15. Indianapolis, IN. [Oral].
- **Vullaganti, N.**[$], Ram, B. G., Zhang, X., Pires, C. B., Alderholdt, W., Overby, P., Sun, X. A novel CNN-based approach for soil nutrient prediction using hyperspectral imaging. 2025 CSABE/ASABE International Meeting, July 13–16. Toronto, ON, Canada. [Oral].
- Zhang, X.[$], Ram, B. G., **Vullaganti, N.**, Alderholdt, W., Overby, P., Sun, X. oil Moisture Prediction using VNIR Spectroscopy and Deep learning for Ground Vehicle Applications. 2025 CSABE/ASABE International Meeting, July 13–16. Toronto, ON, Canada. [Poster].
- **Vullaganti, N.**[$], Rai, N., Sun, X. Applications of Precision Agriculture Technologies in Site-specific Nutrient Management: A Review. 2024 ASABE North Central Regional Section Meeting. April 11 & 12. Brookings, South Dakota. [Poster].

- Attended 2nd Annual North Dakota Contolled Environment Agriculture (CEA) Conference in Bismark, ND, 2025.

Teaching, Mentoring & Outreach

- **Workshop Instructor**, Tech Savvy STEM Event, University of Jamestown, ND, 2026
  Delivered a hands-on workshop on AI, robotics, and UAVs in agriculture to 6th–8th grade students, and guided them in deploying AI models on microcontrollers and robotic platforms for weed detection.
- **Guest Lecturer**, "Introduction to Precision Agriculture", PAG 115, NDSU, Fall 2025
- **Mentor**, Soil Testing Laboratory, Government of Andhra Pradesh, India, Jan–Oct 2023
  Trained and supervised 15 undergraduate interns in soil sampling, nutrient analysis, and laboratory safety.
- **Community engagement**, Rural Awareness Work Experience Program, India, Aug–Dec 2019
  Worked closely with farmers to address challenges in crop production, facilitated knowledge transfer among rural farming communities through farmer training sessions, field demonstrations, and workshops.

Academic Service/Professional Activities

- Reviewer, *Journal of Agriculture and Food Research* (Elsevier), January 2026–present.
- Reviewer, *Data in Brief* (Elsevier), September 2025–present.
- Serving as a judge for MS Student poster competition "Advances in Nutrient Management Poster" at CANVAS 2026, Portland OR.

Leadership & Volunteering

- **Senator, NDSU Student Government, 2025–2026** Represented the College of Agriculture, Food Systems, and Natural Resources in university policy discussions and advocated for student well-being initiatives.
- **Group lead, NDSU Welcome Week, 2025** Mentored a group of 30 incoming undergraduates through orientation and campus integration.
- **Volunteer, 2025 New International Student Orientation, NDSU** Served on the student panel and led discussions to help new international students transition to NDSU.